



From Pixels to Prognosis: Hybrid CNN-Transformer Model for Dementia Detection

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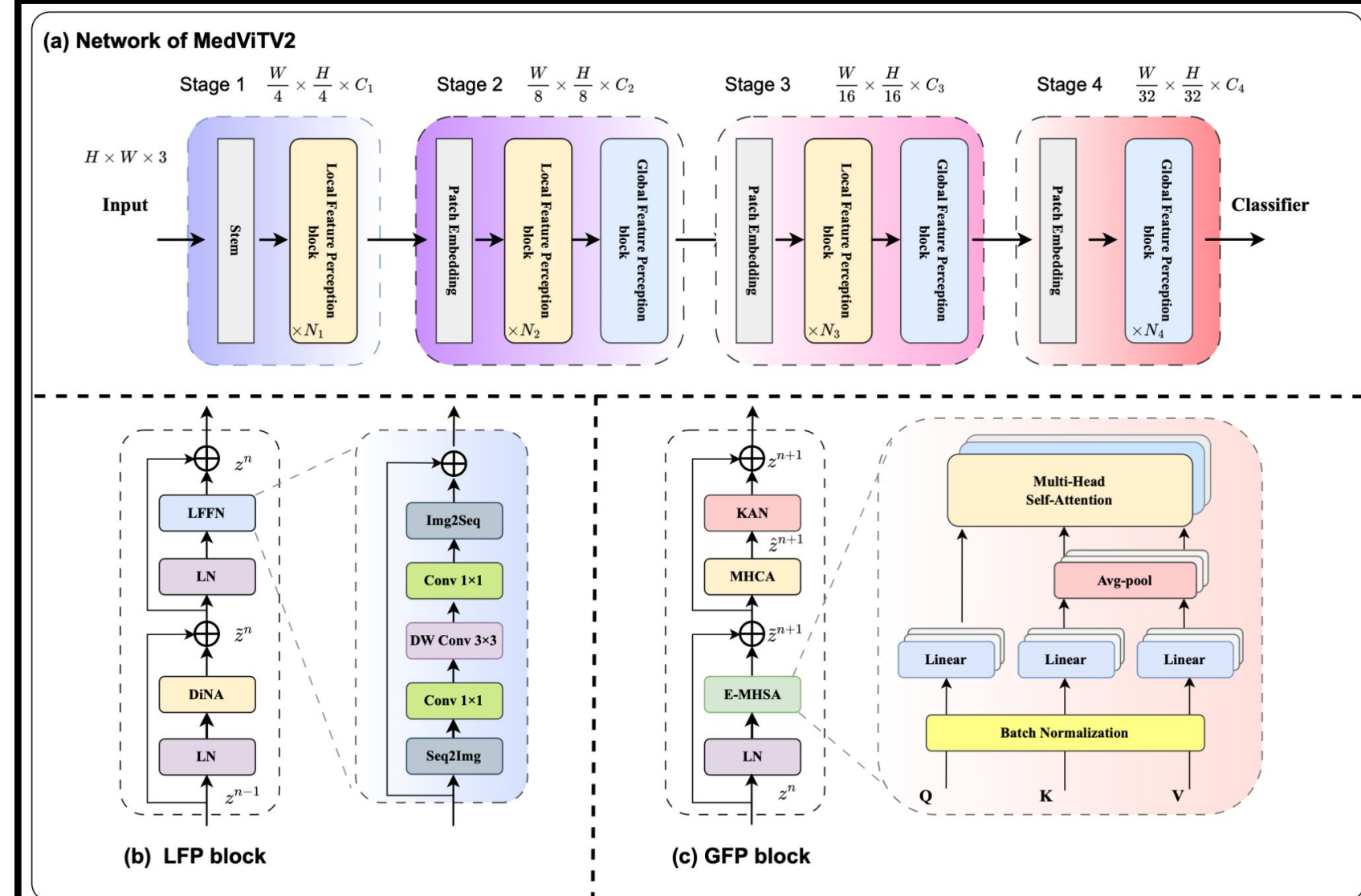


ABSTRACT

MedViT-2 achieves over 99% accuracy on four-class dementia staging from structural MRI, outperforming existing CNN-based baselines on the same dataset. We trained MedViT-2-tiny [2] — a hybrid CNN–Transformer architecture designed for medical imaging but never previously evaluated on neuroimaging — on 44,000 MRI images across four clinical categories. This hybrid architecture captures both local and global patterns critical to dementia severity. These findings highlight the promise of hybrid architectures like MedViT-2 for improving automated dementia detection from structural neuroimaging.

INTRODUCTION

Early, accurate dementia detection is critical for optimizing treatment outcomes. While deep learning has shown promise for automated diagnosis from MRI, conventional CNN-based approaches struggle to model brain-wide patterns. **MedViT-2** addresses this gap through a **hybrid architecture** that jointly encodes **local and global neuroimaging features** for robust dementia staging.



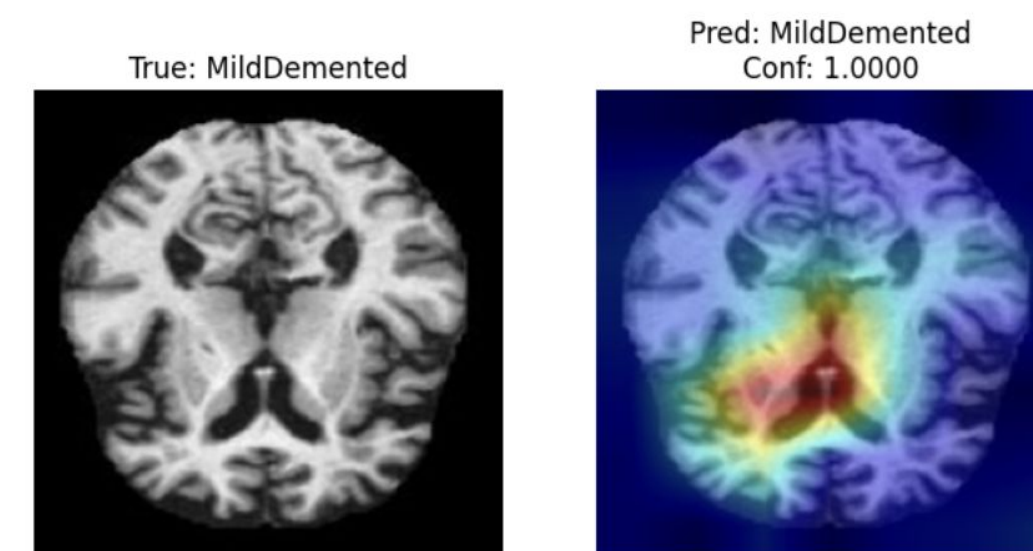
METHODOLOGY

We used an OASIS study-derived structural MRI dataset of 44,000 images [4], labeled into four clinical categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Augmentation and upsampling yielded at least 10,000 images per class, compensating for substantial class imbalance in the original cohort.*

We trained MedViT-2-tiny on the augmented dataset. The model architecture employs **convolutional multi-head attention** to encode long-range dependencies without sacrificing local precision. Training was conducted for 100 epochs and performance was evaluated on a held-out test set – using overall accuracy and class-specific precision, recall, and F1 scores to quantify both global performance and per-class reliability.

*Third-party augmentation was applied to the training data with minimal documentation. To mitigate contamination concerns, results were validated across multiple external dataset with consistent performance.

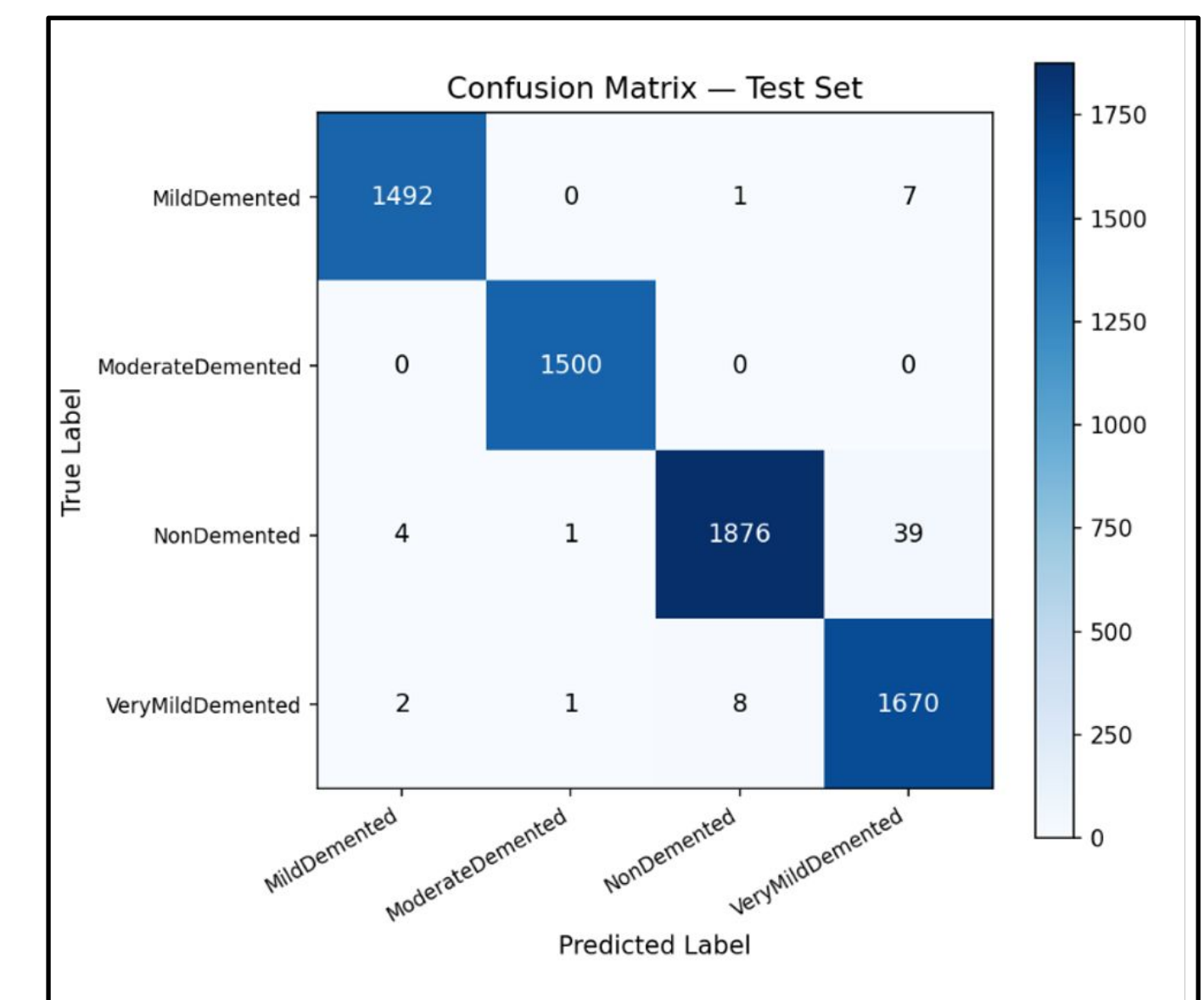
RESULTS



MedViT-2-tiny achieved strong performance on four-class dementia staging. On the held-out test set, overall **accuracy reached 99.1%**, with a weighted **F1-score of 0.9905** and **AUC of 0.997** (OVR). Class-specific F1-scores were similarly high for Mild Demented (0.9953), Moderate Demented (1.0000), Non Demented (0.9868), and Very Mild Demented (0.9832), with per-class specificity above 0.99 for all labels.

CONCLUSION

MedViT-2-tiny demonstrated excellent performance, achieving over 99% overall accuracy and near-perfect class-specific F1-scores. Results outperformed CNN baselines by multiple points, particularly for Very Mild and Mild Demented classes. These findings support the potential of hybrid CNN–Transformer detect dementia **early** and with **improved overall accuracy**. Future work should extend evaluation to include additional neurodegenerative syndromes such as frontotemporal dementia and dementia with Lewy bodies, providing a more robust assessment of MedViT-2 in heterogeneous clinical populations.



ACKNOWLEDGEMENTS

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